

Notes

Chapter 1

1. For documentation of the *Feral Robotic Dogs* project, see <http://www.nyu.edu/projects/xdesign/feralrobots>, accessed November 20, 2010.
2. For an overview of critical design, see Dunne and Raby (2001). For overviews of tactical media, see Garcia and Lovink (1997) and Raley (2009).
3. For example, Web sites such as UNESCO's Design 21: Social Design Network (<http://www.design21sdn.com>), WorldChanging (<http://www.worldchanging.com>), and Design Altruism Project (<http://www.design-altruism-project.org>) chronicle contemporary social design and establish informal and formal networks of resources and practitioners. The +Design program, run by ICOGRADA (International Association of Graphic Design Associations), is an initiative by a professional organization that encourages designers to find design solutions to social problems. (All Web sites were last accessed December 1, 2008.)
4. For key texts by Jürgen Habermas and John Rawls and on deliberative democracy, see Bessette (1994), Elster (1988), Habermas (1989, 1993, 1996a, 1996b), and Rawls (1971, 1993). For a review of these texts and positions in regard to agonistic democracy, see Mouffe (2000a).
5. For an overview of the AIGA *Design for Democracy* projects, see <http://www.aiga.org/design-for-democracy>, accessed January 24, 2010.
6. For documentation of the *Million Dollar Blocks* project, see <http://www.spatialinformationdesignlab.org>, accessed January 24, 2010.
7. The self-published books *The Pattern* and *Architecture and Justice* as well as project documentation for the *Million Dollar Blocks* are available at <http://www.spatialinformationdesignlab.org>, accessed November 20, 2010.
8. See <http://benfry.com/isometricblocks>.
9. See <http://www.dunneandraby.co.uk/content/home>.
10. See <http://www.critical-art.net>.
11. See <http://www.appliedautonomy.com/isee.html>.
12. For overviews of political posters and other forms of political graphic design, see Glaser and Ilic (2006), Lasn (2006), and McQuiston (1995). For an overview of the work of Krzysztof Wodiczko, see Wodiczko (1999).

13. For overviews on Constructivism and the Bauhaus, see Fiedler and Feierabend (2008), Frampton (2007), Gropius (1965), James-Chakraborty (2006), Margolin (1998), and Rickey (1995).

Chapter 2

1. *State-Machine: Agency* is accessible at <http://state-machine.org/agency>, accessed August 22, 2011.

2. The *Naming Names* visualization is available at <http://www.nytimes.com/interactive/2007/12/15/us/politics/DEBATE.html>, accessed August 22, 2011.

3. For a discussion of visualizations as a form of journalism, see Bogost, Ferrari, and Schweizer (2010).

4. *The Dumpster* is available at <http://artport.whitney.org/commissions/thedumpster>, accessed August 22, 2011.

5. *We Feel Fine* is available at <http://www.wefeelfine.org>, accessed August 22, 2011.

6. The work of the design studio Stamen provides exceptional examples of interactive map products and services that weave together spatial data, photographs, and user-generated content and demonstrate the power of transcoding. In 2010, Stamen released two products *Polymaps* and *Prettymaps*, both of which enable designers and end-users to construct visually rich maps that integrate and display multiple forms of media content. See <http://www.stamen.com>, accessed August 22, 2011.

7. The CIA's *World Factbook* is available at <https://www.cia.gov/library/publications/the-world-factbook>, accessed August 22, 2011.

8. The *Greenpeace Blacklist* is available at <http://www.greenpeace.org/international/en/campaigns/oceans/pirate-fishing/Blacklist1>, accessed August 22, 2011.

9. For a comprehensive list of social network analysis software, see <http://www.insna.org/software/index.html>, maintained by the International Networks for Social Network Analysis, accessed August 22, 2011. For an overview of the technical approaches and issues to social network analysis, see Brandes and Erlebach (2005).

10. *They Rule* is available at <http://www.theyrule.net>, accessed August 22, 2011.

11. *Exxon Secrets* is available at <http://www.exxonsecrets.org/maps.php>, accessed August 22, 2011.

12. The music mashup is an extension of the remix toward something that is more like a composite. It still elides melding to the point of not recognizing or being able to separate the source materials, which characterizes the composite. One appealing characteristic of the music mashup is that it allows the recognition of the appropriation and application of a beat or lyric from one musician to another or strung together to make an entirely new third musical statement. Just as the remix was enabled with tools such as turntables, mixers, and tape decks, the mashup was enabled with digital audio tools. Software packages made possible two basic operations—beat matching between two songs, which allows the tempos to be integrated, and separating vocal from instrumental tracks through software equalizing. A simple recipe for making a mashup is to take one artist's track, remove the vocal, and then

place another artist's vocal on top of the first artist's track. Although the mashup scene has a rich and full history, it received popular attention with the 2004 release of *The Grey Album* by Danger Mouse, which layered a cappella vocals from rap artist Jay-Z's *The Black Album* over tracks from the Beatles' *The White Album* to produce a novel, conceptually compelling, aesthetically pleasing, and completely unauthorized and therefore illegal new hybrid album, which received critical acclaim as well as a cease-and-desist letter from EMI, holders of the copyright.

13. *Unfluence* is available at <http://unfluence.primite.net>, accessed November 19, 2011.

14. The Sunlight Foundation is a nonprofit organization that promotes government transparency and accountability by providing access to government-related data and application programming interfaces (APIs) for using that data.

15. Although the term *extension* can refer to any kind software application, it is most commonly used in reference to Web browser software.

16. *MAICgregator* is available at <http://maicgregator.org>, accessed August 22, 2011.

17. See <http://maicgregator.org/FAQ>, accessed August 22, 2011.

18. *Oil Standard* is available at <http://turbulence.org/Works/oilstandard>, accessed August 22, 2011.

19. For a discussion of how games operate only by playing extensions and how they relate to computational media more generally, see Bogost (2007).

20. *MAICgregator* does allow users to adjust the settings to undo that quality of seamlessness.

Chapter 3

1. See Herbert Simon (1996) as an example of a canonical text of classical artificial intelligence, which argues for intelligence as symbol manipulation. Debates concerning the relationship between symbol manipulation and computational intelligence arose early on in AI through the work of Hubert Dreyfus (1972) and later John Haugeland (1985) and were continued into 1990s in the field of science studies by Lucy Suchman (1987). For an overview of "nouvelle AI," see Rodney Brooks (1999).

2. For documentation of PARO, see <http://www.parorobots.com>, accessed October 18, 2010.

3. For a discussion of what the engineers and designers mean by "mental commitment robot," see <http://www.paro.jp/english/about.html>, accessed October 18, 2010.

4. This general theme is repeated throughout the research and marketing literature on PARO produced by the engineers and scientists at the National Institute of Advanced Industrial Science and Technology. For an argument of PARO as animal-like therapy, see Kaoru Inoue, Kazuyoshi Wada, and Yuku Ito (2008).

5. For a listing of publications on PARO, see <http://www.parorobots.com/whitepapers.asp>, accessed October 18, 2010.

6. *Ibid.*

7. For an overview of the marketing and public positioning of PARO, see <http://www.parorobots.com>, accessed October 18, 2010.
8. For an in-depth discussion, see Sherry Turkle (2011).
9. Phenomenology is a diverse field, and the treatment of embodiment is rigorous and nuanced, but in the phenomenological sense, embodiment refers to the fact that an entity has a body, and from that starting position, philosophical inquiries are made into the relations between the qualities of that body and a given entity's worldly capabilities and experience. In phenomenology, experience and the body are both sites of inquiry where terms and meanings are meted out. For example, for Maurice Merleau-Ponty (1962), the emphasis was not on a generic physiological body but rather on what he termed "the phenomenal body," which is a distinctive, individualized body that directs our relations with others (people and objects) in the world. For a discussion of phenomenology in relation to computational media and embodiment, see Paul Dourish (2001). For overviews of embodied cognition, see Andy Clark (1997), George Lakoff and Mark Johnson (1999), and Margaret Wilson (2002).
10. See Lucy Suchman's discussion of the work of the artist Stelarc (Suchman 2006, 241–258) and John Johnston (2008).
11. For documentation of *Blendie*, see Kelly Dobson (2007a) and <http://web.media.mit.edu/~monster/blendie>, accessed October 18, 2010.
12. Ibid.
13. Although Ernst Jentsch first discussed the notion of the uncanny in his 1906 essay "On the Psychology of the Uncanny," Sigmund Freud's essay is the more common source for an explanation of the uncanny.
14. This film is a remake of *Chakushin Ari* (2003).
15. Given the limitations of robotics technology and the constraints of laboratory-based studies, research on the uncanny valley is challenging because it is difficult to construct a robot and a situation where an experience of the uncanny might be measured in a scientific manner. Nonetheless, within the field of human-robot interaction, there is a growing body of research on the topic. Examples include Michael L. Walters and colleagues (2008) and Chin-Chang Ho, Karl L. MacDorman, and Z. A. D. Dwi Pramono (2008).
16. For an overview of this move away from a singular focus on productivity and toward new themes in information and communication technology design, see William J. Mitchell, Alan S. Inouye, and Marjory S. Blumenthal (2003).
17. As specific examples, see Phoebe Sengers, Kirsten Boehner, Shay David, and Joseph Kaye (2005) and Sengers and Bill Gaver (2006).
18. For an overview of PaPeRo, see <http://www.nec.co.jp/products/robot/en/index.html>, accessed October 18, 2010.
19. For a detailed discussion of the design of *Omo*, see Dobson (2007a, 98–121).
20. For documentation of *Amy and Klara*, see <http://www.realtechsupport.org/repository/male-dicta.html>, accessed October 18, 2010.
21. Suchman (2006, 241–258) makes a similar observation concerning her encounter with the artist Stelarc's *Head*, which is a system consisting of a larger-than-life

projection of a three-dimensional virtual model of the artist's head and chat software that enables conversation between the prosthetic head and gallery visitors.

22. For in-depth descriptions of the text-to-speech software and the process of computationally constructing accents in *Amy and Klara*, see Böhlen (2006a, 2006b, 2008).

23. In this way, speech becomes available for what Bogost (2006, 3) calls "unit operations" or "modes of meaning-making that privilege discrete, disconnected actions."

Chapter 4

1. For documentation of the *Spore 1.1* project, see <http://swamp.nu/projects/spore1>, accessed November 22, 2010.

2. Although in 1991 this was a novel idea, today phones contain stunning computational capacities, and the idea of imbuing computation into objects other than the traditional computer seems obvious. But the vision of ubicomp espoused by Weiser and most contemporary researchers extends beyond smart phones by imbuing computation into all types of objects and environments.

3. In the project *Spore 1.1*, rather than having the computer disappear, its design explicitly brings the computer into view. The tree could have been housed in a large pot with opaque sides that would have kept the wires, fans, and circuit boards out of view. That much computer stuff is no longer needed to provide that functionality. A set of circuit boards that would occupy one-fifth the amount of space could now easily provide the same functionality. But the design of *Spore 1.1* is, or at least was, attention-grabbing because it showed a working vision of the near future that is both exploratory and explanatory of this notion of combing computation with everyday objects.

4. For marketing materials describing the Ambient Umbrella, see <http://www.ambientdevices.com/products/umbrella.html>, accessed November 22, 2010.

5. Consider the redesign of the British Petroleum logo. By using the glyph of the Greek *helios*, or sun, and a linguistic switch, in which *bp* is used as a stand-in for the phrase "Beyond Petroleum," the redesigned logo draws from and establishes questionable associations with environmentalism.

6. Many of these carbon footprint tools are related to advocacy. For example, the Nature Conservancy offers an online carbon calculator at <http://www.nature.org/greenliving/carboncalculator>. A similar calculator is offered by the United States Environmental Protection Agency at http://www.epa.gov/climatechange/emissions/ind_calculator.html. These calculators and countless others lead a user through a survey, collecting information on a range of topics from home heating and cooling to food and diet. From this information, the calculators generate both a carbon footprint record as well as suggestions for possible changes in behavior to lessen that footprint. All URLs accessed November 22, 2010.

7. For documentation of the *Natural Fuse* project, see <http://www.haque.co.uk/naturalfuse.php>, accessed November 22, 2010.

8. The value of the carbon offset is provided as a constant in the software—that is, it is a predetermined value.
9. For documentation of the *Ad-hoc Dark (roast) Network Travel Mug*, see http://survival.sentientcity.net/blog/?page_id=22, accessed November 22, 2010.
10. For discussions of the effects of contemporary technology on the changing character of cities, see Manuel Castells (2009), Stephen Graham and Simon Marvin (2001), and Saskia Sassen (2002).
11. For a collection of essays on urban informatics and urban computing, see Marcus Foth (2009).
12. For marketing materials describing the ShotSpotter Gunshot Location System, see <http://www.shotspotter.com>, accessed November 22, 2010.
13. For documentation of the *CCD-Me Not Umbrella*, see http://survival.sentientcity.net/blog/?page_id=17, accessed November 22, 2010.

Chapter 5

1. This does not mean that one would always go from information design to robots to ubiquitous computing. The tactics of adversarial design are not bound to any one category of objects, to any one quality, or even to the medium of computation. Each tactic was set with a genre and quality to create a resonant pairing that amplifies and gives clarity to the relations and affects, but they are not fixed to one another or mutually exclusive.